

Fermat's last theorem for $n=4$
(any field K has been replaced with $\mathbb{Q}$)

Def * Elliptic curve over $\mathbb{Q}$ is equation of form:
 $y^2 = ax^3 + bx^2 + cx + d$ ($a, b, c, d \in \mathbb{Q}$, $a \neq 0$)
 where RHS polynomial has no multiple root

Def For $a = \frac{m}{n}$ with $m, n \in \mathbb{N}$, $(m, n) = 1$
 define height $H(a) := \max(|m|, |n|)$

(x,y) on elliptic curve
 $\Rightarrow (\frac{-1}{x}, \frac{y}{x^2})$ on as well,
 $H(x)=H(-1/x)$

$$\begin{aligned} i:(x,y,z) &\mapsto \left(\frac{y-x}{2}, \frac{z}{2}, \frac{x+y}{2}\right) \\ j:(u,v,w) &\mapsto (w-u, w+u, 2v) \end{aligned}$$

Lemma 1.5 For distinct $a, b, c \in \mathbb{Q}$ define B , $\tilde{C}$ and C

$$B = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + a = v^2 + b = w^2 + c\}$$

$$\tilde{C} = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c)\}.$$

$$C = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c), y \neq 0\}.$$

(1) $\exists$ mutually inverse maps $f: B \rightarrow C$, $g: C \rightarrow B$

$$f(u, v, w) = (u^2 + a + uv + vw + wu, (u+v)(v+w)(w+u),$$

$$g(x, y) = \left(\frac{1}{2y}((x-a)^2 - (b-a)(c-a)), \right.$$

$$\left. \frac{1}{2y}((x-b)^2 - (a-b)(c-b)), \frac{1}{2y}((x-c)^2 - (a-c)(b-c)) \right)$$

(2) $\exists$ map $h: B \rightarrow \tilde{C}$, $h(u, v, w) = (u^2 + a, uvw)$

(i) Let (x_0, y_0) on $y^2 = x^3 - x$ with $y_0 \neq 0$ and minimal $H(x_0)$; assume $1 < x_0$

$$\begin{cases} i:A_d \rightarrow B_d \\ j:B_d \rightarrow A_d \end{cases}$$
$$a=d, b=0, c=-d, \begin{cases} l:B_d \rightarrow C_d \\ g:C_d \rightarrow B_d \end{cases}$$

Lem 1.4 For $d \in \mathbb{Q}$ define:
 $A_d = \{(x, y, z) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid x^2 + y^2 = z^2, \frac{1}{2}xy = d\}$
 $B_d = \{(u, v, w) \in \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \mid u^2 + d = v^2, v^2 + d = w^2\}$
 $C_d = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = x^3 - d^2x, y \neq 0\}$.
 $\exists$ bijections between any 2 of A_d, B_d, C_d

Rem 1.6 $h \circ g: C \rightarrow \tilde{C}$ is called multiplication-by-2 map for elliptic curve $y^2 = (x-a)(x-b)(x-c)$; image of $h \circ g$ is $\{(x,y) \in \mathbb{Q} \times \mathbb{Q} \mid y^2 = (x-a)(x-b)(x-c), x-a, x-b, x-c \text{ are } \square \text{ in } \mathbb{Q}\}$ because:
 $(x',y') \in h(u,v,w) = x' = u^2 + a \Rightarrow x'-a \text{ is } \square \text{ in } \mathbb{Q}$
 from definition of B (image of g): $u^2 + a = v^2 + b = w^2 + c$
 $x'-b = (v^2 + b) - b = v^2 \text{ is } \square \text{ in } \mathbb{Q}$
 $x'-c = (w^2 + c) - c = w^2 \text{ is } \square \text{ in } \mathbb{Q}$

$$a=1, b=0, c=-1$$
$$\left\{ \begin{array}{ll} \text{(i)} & \Rightarrow A_d \neq \emptyset \\ \text{(ii)} & \Rightarrow B_d \neq \emptyset \\ \text{(iii)} & \Rightarrow C_d \neq \emptyset \end{array} \right.$$

Lem 1.3 $d \in \mathbb{Q}$, these are equivalent:

- (i) $\exists$ rational number sides **right** $\triangle$ of area d
- (ii) $\exists$ 3 $\square$ s of rational numbers that form arithmetic progression of difference d
- (iii) $\exists$ rational solution to $y^2 = x^3 - d^2x$ other than $(x,y) = (0,0)$ and $(\pm d, 0)$

Prop 0.13 The area of a right $\triangle$ whose sides are integers is not a $\square$

$$\neg(\text{iii})$$

Prop 1.2 The only rational solutions to $y^2=x^3-x$ are $(x,y)=(0,0)$ and $(\pm 1,0)$

$$1.1 \text{ sol} \cdot z^2/y^6, \text{ 4:}$$

$$0 \neq \left(\frac{x^2z}{y^3}\right)^2 = \left(\frac{z^2}{y^2}\right)^3 - \frac{z^2}{y^2}$$

Prop 1.1 $\nexists$ solution (x,y,z) to $x^4+y^4=z^4$ satisfying $xyz \neq 0$

(ii) x_0-1, x_0 and x_0+1 are all squares
 $\Rightarrow \exists (x_1, y_1) \in C: h \circ g(x_1, y_1) = (x_0, y_0)$

(iii) $H(x_1) < H(x_0)$, ∇